A FINITE TYPE SEPARATED SCHEME COVERED BY TWO OPENS WITH COMPACTIFICATIONS AND DENSE INTERSECTION HAS A COMPACTIFICATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0F40 (sources/stacks-flat/pinned-original-source.tex line 9791).

Allowed standard prerequisites (editorial).

- Compactifications of the pieces and a proper morphism dominating U. Same chapter, Lemma labelled lemma-compactification-dominates, cited at proof lines 9801-9803. Provides the V_i -> U proper morphisms used to control the closures of the Z_i.
- Separating disjoint locally closed subschemes by an admissible blowing up, and making a proper morphism an isomorphism along a closed subscheme. Same chapter, Lemmas labelled lemma-separate-disjoint-locally-closed-by-blowing-up and lemma-blowup-iso-along, cited at proof lines 9828-9830 and 9856-9858. Produces the admissible blowups that make the closures disjoint and p_1 an isomorphism near the closure.
- Strict transforms under admissible blowups and the glued separatedness criterion. Same chapter, Lemmas labelled lemma-replaced-by-strict-transform and lemma-check-separated, cited at proof lines 9836-9838 and 9870-9871. Tracks X_{12} under the blowups and verifies that W_i -> S is separated.
- The refined valuative criterion for universal closedness and a common blowup diagram. Morphisms, Lemma labelled morphisms-lemma-refined-valuative-criterion-universally-closed, cited at proof lines 9943-9945; Same chapter, Lemma labelled lemma-find-common-blowups, cited at proof line 9907; Morphisms, Lemma labelled morphisms-lemma-characterize-proper, cited at proof lines 9894-9896. Converts the extension property for valuation rings into universal closedness of W -> S.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0F3Z: sources/stacks-flat/pinned-original-source.tex, source lines 9762–9762. Same-chapter pointer into the author environment lemma at source line 9762; invoked at proof line 9803 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F3V: sources/stacks-flat/pinned-original-source.tex, source lines 9554–9554. Same-chapter pointer into the author environment lemma at source line 9554; invoked at proof line 9831 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01KR: sources/stacks-chow/tags.txt, source lines 2060–2060. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9840 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F3Y: sources/stacks-flat/pinned-original-source.tex, source lines 9728–9728. Same-chapter pointer into the author environment lemma at source line 9728; invoked at proof line 9852 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F3W: sources/stacks-flat/pinned-original-source.tex, source lines 9608–9608. Same-chapter pointer into the author environment lemma at source line 9608; invoked at proof line 9868 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F3U: sources/stacks-flat/pinned-original-source.tex, source lines 9540–9540. Same-chapter pointer into the author environment lemma at source line 9540; invoked at proof line 9896 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BX5: sources/stacks-chow/tags.txt, source lines 14968–14968. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9915 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F3X: sources/stacks-flat/pinned-original-source.tex, source lines 9647–9647. Same-chapter pointer into the author environment lemma at source line 9647; invoked at proof line 9929 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0894: sources/stacks-chow/tags.txt, source lines 10484–10484. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9945 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let S be a Noetherian scheme. Let U be a scheme of finite type and separated over S. Let $U = U_1 \cup U_2$ be opens such that U_1 and U_2 have compactifications over S and such that $U_1 \cap U_2$ is dense in U. Then U has a compactification over S.*

Proof. Choose a compactification $U_i \subset X_i$ for $i = 1, 2$. We may assume U_i is scheme theoretically dense in X_i. We may assume there is an open $V_i \subset X_i$ and a

proper morphism $\psi_i : V_i \to U$ extending $\mathrm{id} : U_i \to U_i$, see Lemma 0F3Z. Picture

$$\begin{array}{ccccc} U_i & \longrightarrow & V_i & \longrightarrow & X_i \\ \downarrow & \swarrow_{\psi_i} & & & \\ U & & & & \end{array}$$

If $\{i, j\} = \{1, 2\}$ denote $Z_i = U \setminus U_j = U_i \setminus (U_1 \cap U_2)$ and $Z_j = U \setminus U_i = U_j \setminus (U_1 \cap U_2)$. Thus we have

$$U = U_1 \amalg Z_2 = Z_1 \amalg U_2 = Z_1 \amalg (U_1 \cap U_2) \amalg Z_2$$

Denote $Z_{i,i} \subset V_i$ the inverse image of Z_i under ψ_i. Observe that ψ_i is an isomorphism over an open neighbourhood of Z_i. Denote $Z_{i,j} \subset V_i$ the inverse image of Z_j under ψ_i. Observe that $\psi_i : Z_{i,j} \to Z_j$ is a proper morphism. Since Z_i and Z_j are disjoint closed subsets of U, we see that $Z_{i,i}$ and $Z_{i,j}$ are disjoint closed subsets of V_i.

Denote $\overline{Z}_{i,i}$ and $\overline{Z}_{i,j}$ the closures of $Z_{i,i}$ and $Z_{i,j}$ in X_i. After replacing X_i by a V_i-admissible blowup we may assume that $\overline{Z}_{i,i}$ and $\overline{Z}_{i,j}$ are disjoint, see Lemma 0F3V. We assume this holds for both X_1 and X_2. Observe that this property is preserved if we replace X_i by a further V_i-admissible blowup.

Set $V_{12} = V_1 \times_U V_2$. We have an immersion $V_{12} \to X_1 \times_S X_2$ which is the composition of the closed immersion $V_{12} = V_1 \times_U V_2 \to V_1 \times_S V_2$ (Schemes, Lemma 01KR) and the open immersion $V_1 \times_S V_2 \to X_1 \times_S X_2$. Let $X_{12} \subset X_1 \times_S X_2$ be the scheme theoretic image of $V_{12} \to X_1 \times_S X_2$. The projection morphisms

$$p_1 : X_{12} \to X_1 \quad \text{and} \quad p_2 : X_{12} \to X_2$$

are proper as X_1 and X_2 are proper over S. If we replace X_1 by a V_1-admissible blowing up, then X_{12} is replaced by the strict transform with respect to this blowing up, see Lemma 0F3Y.

Denote $\psi : V_{12} \to U$ the compositions $\psi = \psi_1 \circ p_1|_{V_{12}} = \psi_2 \circ p_2|_{V_{12}}$. Consider the closed subscheme

$$Z_{12,2} = (p_1|_{V_{12}})^{-1}(Z_{1,2}) = (p_2|_{V_{12}})^{-1}(Z_{2,2}) = \psi^{-1}(Z_2) \subset V_{12}$$

The morphism $p_1|_{V_{12}} : V_{12} \to V_1$ is an isomorphism over an open neighbourhood of $Z_{1,2}$ because $\psi_2 : V_2 \to U$ is an isomorphism over an open neighbourhood of Z_2 and $V_{12} = V_1 \times_U V_2$. By Lemma 0F3W there exists a V_1-admissible blowing up $X_1' \to X_1$ such that the strict transform $p_1' : X_{12}' \to X_1'$ of p_1 is an isomorphism over an open neighbourhood of the closure of $Z_{1,2}$ in X_1'. After replacing X_1 by X_1' and X_{12} by X_{12}' we may assume that p_1 is an isomorphism over an open neighbourhood of $\overline{Z}_{1,2}$.

The reduction of the previous paragraph tells us that

$$X_{12} \cap (\overline{Z}_{1,2} \times_S \overline{Z}_{2,1}) = \emptyset$$

where the intersection taken in $X_1 \times_S X_2$. Namely, the inverse image $p_1^{-1}(\overline{Z}_{1,2})$ in X_{12} maps isomorphically to $\overline{Z}_{1,2}$. In particular, we see that $Z_{12,2}$ is dense in $p_1^{-1}(\overline{Z}_{1,2})$. Thus p_2 maps $p_1^{-1}(\overline{Z}_{1,2})$ into $\overline{Z}_{2,2}$. Since $\overline{Z}_{2,2} \cap \overline{Z}_{2,1} = \emptyset$ we conclude.

Consider the schemes

$$W_i = U \coprod\nolimits_{U_i} (X_i \setminus \overline{Z}_{i,j}), \quad i = 1, 2$$

obtained by glueing. Let us apply Lemma 0F3U to see that $W_i \to S$ is separated. First, $U \to S$ and $X_i \to S$ are separated. The immersion $U_i \to U \times_S (X_i \setminus \overline{Z}_{i,j})$ is closed because any specialization $u_i \rightsquigarrow u$ with $u_i \in U_i$ and $u \in U \setminus U_i$ can be lifted uniquely to a specialization $u_i \rightsquigarrow v_i$ in V_i along the proper morphism $\psi_i : V_i \to U$

and then v_i must be in $Z_{i,j}$. Thus the image of the immersion is closed, whence the immersion is a closed immersion.

On the other hand, for any valuation ring A over S with fraction field K and any morphism $\gamma : \mathrm{Spec}(K) \to (U_1 \cap U_2)$ over S, there is an i and an extension of γ to a morphism $h_i : \mathrm{Spec}(A) \to W_i$. Namely, for both $i = 1, 2$ there is a morphism $g_i : \mathrm{Spec}(A) \to X_i$ extending γ by the valuative criterion of properness for X_i over S, see Morphisms, Lemma 0BX5. Thus we only are in trouble if $g_i(\mathfrak{m}_A) \in \overline{Z}_{i,j}$ for $i = 1, 2$. This is impossible by the emptyness of the intersection of X_{12} and $\overline{Z}_{1,2} \times_S \overline{Z}_{2,1}$ we proved above.

Consider a diagram

$$\begin{array}{ccccc} W_1' & \longrightarrow & W & \longleftarrow & W_2' \\ \downarrow & \nwarrow & \uparrow & \nearrow & \downarrow \\ W_1 & \longleftarrow & U & \longrightarrow & W_2 \end{array}$$

as in Lemma 0F3X. By the previous paragraph for every solid diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \xrightarrow{\gamma} & W \\ \downarrow & \nearrow & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & S \end{array}$$

where $\mathrm{Im}(\gamma) \subset U_1 \cap U_2$ there is an i and an extension $h_i : \mathrm{Spec}(A) \to W_i$ of γ. Using the valuative criterion of properness for $W_i' \to W_i$, we can then lift h_i to $h_i' : \mathrm{Spec}(A) \to W_i'$. Hence the dotted arrow in the diagram exists. Since W is separated over S, we see that the arrow is unique as well. This implies that $W \to S$ is universally closed by Morphisms, Lemma 0894. As $W \to S$ is already of finite type and separated, we win. $\square$